

Statistical Estimation via OLS and MLE

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- Dataset collected in Fall 2025
- Source: BoardGameGeek XML API Documentation
- Additional reference: BGG API Overview
- Processing code available on GitHub Repository
- Target variable: Bayesian average rating (Y)
- Predictors: year published, complexity, playing time, player characteristics
- Final sample size: 3,833 games (minimum 1000 ratings)

Estimate the relationship between game characteristics
and average rating

Using a linear regression framework grounded in
Maximum Likelihood Estimation

$$Y_i = \beta_0 + \beta_1 X_{i1} + \cdots + \beta_k X_{ik} + \varepsilon_i$$

$$\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$$

Under the normality assumption:

$$\hat{\beta} = \arg \max_{\beta} L(\beta)$$

Maximizing the likelihood is equivalent to minimizing squared error.

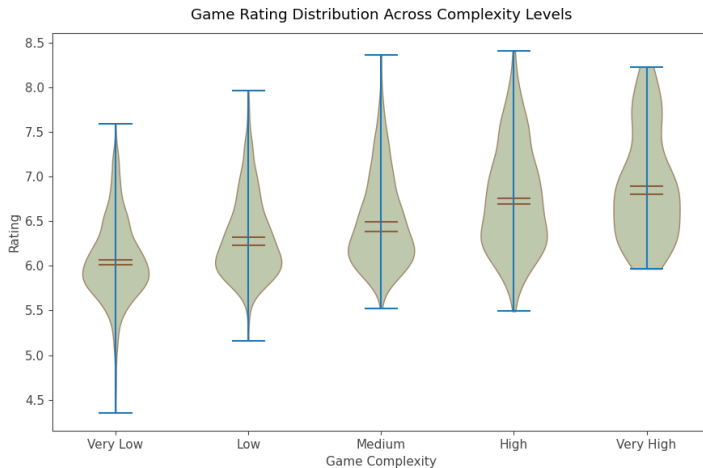
$\Rightarrow$ **Ordinary Least Squares**

Thus, OLS is the MLE estimator under Gaussian errors.

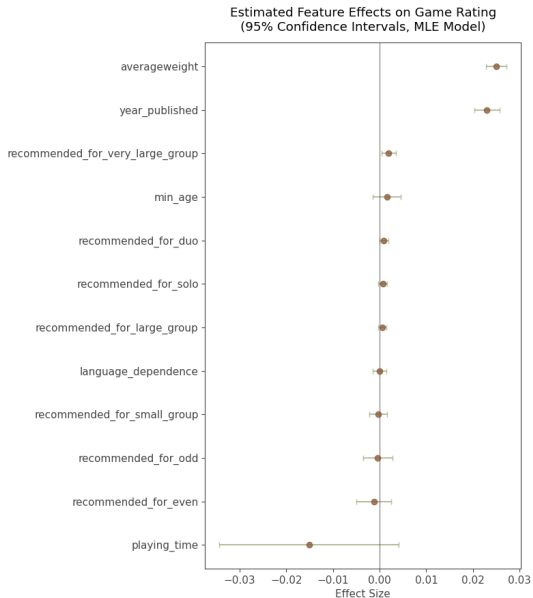
```
print(results.summary())
```

This implements Ordinary Least Squares, which corresponds to the MLE solution under normal errors.

Pre-Regression Distribution



OLS Fit



Residual Distribution

